

5 **(long)** A crane lifts a mass of 4 kg at a speed of 1 ms^{-1} . The crane's 50 Hz AC motor can be modelled as a 50 mH inductance in series with a resistor R . The resistor dissipates a power equal to the output of the motor. State any assumptions made when answering the following questions:

(a) If the input current is 2 A when lifting this load, what is the power factor of the circuit? [8]

(b) If the crane is driven by an ac supply with a higher voltage through an ideal step-down transformer with a turns ratio of 20:1, what capacitance can be placed across the transformer's high voltage terminals to give the circuit a power factor of unity? [12]

(c) If instead of needing power factor correction, the transformer's winding losses for this load are known to be 3 W and 4 VARs, what will be the input high voltage and the power factor? [10]